

# FIRST BLASTp SEARCH

## 1. SEARCH

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Standard Protein BLAST

BLASTP programs search protein databases using a protein query. more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) Query subrange [?](#)

AAA59172 From  To

Or, upload file  No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database  [?](#)

Organism  [Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

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## 2. RESULTS

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Job Title AAA59172:insulin [Homo sapiens]

RID [674KGFHM014](#) Search expires on 07-26 00:27 am [Download All](#) [?](#)

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Database [ClusteredNR](#) [See details](#) [?](#)

Query ID [AAA59172.1](#)

Description insulin [Homo sapiens]

Molecule type amino acid

Query Length 110

Other reports [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#) [?](#)

**Clusters** [Graphic Summary](#) [Alignments](#) [Taxonomy](#)

Clusters producing significant alignments

Download [?](#) Select columns [?](#) Show 100 [?](#)

**Filter Results**

Organism only top 20 will appear

Type common name, binomial, taxid or group name

[+ Add organism](#)

Percent Identity  to  E value  to  Query Coverage  to

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| Clusters                                                                |                             |                                                         |           |             |             |         |            |          |                |
|-------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------|-----------|-------------|-------------|---------|------------|----------|----------------|
| Graphic Summary                                                         |                             |                                                         |           |             |             |         |            |          |                |
| Alignments                                                              |                             |                                                         |           |             |             |         |            |          |                |
| Taxonomy                                                                |                             |                                                         |           |             |             |         |            |          |                |
| Clusters producing significant alignments                               |                             |                                                         |           |             |             |         |            |          |                |
| Download Select columns Show 100 ?                                      |                             |                                                         |           |             |             |         |            |          |                |
| select all 100 clusters selected                                        |                             |                                                         |           |             |             |         |            |          |                |
| GenPept Graphics Distance tree of results Multiple alignment MSA Viewer |                             |                                                         |           |             |             |         |            |          |                |
| Cluster Composition                                                     | Cluster Ancestor            | Cluster Representative Sequence                         | Max Score | Total Score | Query Cover | E value | Per. Ident | Acc. Len | Accession      |
| Click the icon to see the cluster contents                              |                             |                                                         |           |             |             |         |            |          |                |
| 17 member(s), 11 organism(s)                                            | primates                    | insulin preproprotein [Pan troglodytes]                 | 221       | 221         | 100%        | 1e-72   | 98.18%     | 110      | NP_001008996.1 |
| 3 member(s), 3 organism(s)                                              | primates                    | insulin [Pan paniscus]                                  | 224       | 224         | 100%        | 2e-72   | 99.09%     | 204      | XP_034787832.1 |
| 5 member(s), 5 organism(s)                                              | primates                    | insulin isoform X1 [Macaca nemestrina]                  | 221       | 221         | 100%        | 4e-72   | 98.18%     | 153      | XP_011719619.2 |
| 1 member(s), 1 organism(s)                                              | green monkey                | insulin isoform X2 [Chlorocebus sabaeus]                | 218       | 218         | 100%        | 5e-71   | 97.27%     | 144      | XP_008002825.1 |
| 1 member(s), 1 organism(s)                                              | black snub-nosed monkey     | insulin isoform X2 [Rhinopithecus bieti]                | 218       | 218         | 100%        | 1e-70   | 97.27%     | 147      | XP_017743290.1 |
| 1 member(s), 1 organism(s)                                              | green monkey                | insulin isoform X1 [Chlorocebus sabaeus]                | 218       | 218         | 100%        | 7e-70   | 97.27%     | 209      | XP_008002752.1 |
| 5 member(s), 5 organism(s)                                              | primates                    | INS isoform 5 [Pongo abelii]                            | 191       | 191         | 100%        | 1e-60   | 88.18%     | 98       | PN116423.1     |
| 2 member(s), 2 organism(s)                                              | placental                   | insulin [Tupaia tupaia]                                 | 184       | 184         | 100%        | 4e-58   | 90.00%     | 110      | KAN7991649.1   |
| 1 member(s), 1 organism(s)                                              | European harvest mouse      | insulin [Micromys minutus]                              | 184       | 184         | 100%        | 5e-58   | 79.09%     | 110      | KAN3685427.1   |
| 1 member(s), 1 organism(s)                                              | other sequences             | ZZ-proinsulin [Cloning vector pET28a_ZZ_proinsulin]     | 183       | 183         | 78%         | 3e-56   | 100.00%    | 210      | UFG08408.1     |
| 4 member(s), 4 organism(s)                                              | bats                        | insulin [Rhinolophus affinis]                           | 178       | 178         | 100%        | 2e-55   | 88.18%     | 110      | KAN3685427.1   |
| 1 member(s), 1 organism(s)                                              | high G+C Gram-positive b... | hypothetical protein [Streptomyces chartreusis]         | 179       | 179         | 78%         | 2e-55   | 100.00%    | 143      | MCZ4612290.1   |
| 2 member(s), 1 organism(s)                                              | other sequences             | HPPI [synthetic construct]                              | 175       | 175         | 78%         | 1e-54   | 98.84%     | 97       | AGG11869.1     |
| member(s), organism(s)                                                  | Rhesus monkey               | insulin pro- [Macaca mulatta]                           | 174       | 174         | 78%         | 3e-54   | 98.84%     | 86       | 1006230A       |
| 1 member(s), 1 organism(s)                                              | reed vole                   | insulin [Alexandromys fortis]                           | 175       | 175         | 100%        | 5e-54   | 85.45%     | 128      | XP_049992438.1 |
| 1 member(s), 1 organism(s)                                              | Aston's brown mouse         | hypothetical protein STEG23_014591 [Scotinomys teguina] | 175       | 175         | 100%        | 9e-54   | 85.45%     | 145      | KAL6031003.1   |

Clusters

Graphic Summary

Alignments

Taxonomy

Alignment view

Pairwise

2

Restore defaults

Download

100 clusters selected

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GenPept

Graphics

insulin preproprotein [Pan troglodytes]

Sequence ID: NP\_001008996.1 Length: 110 Number of Matches: 1

Range 1: 1 to 110

GenPept

Graphics

▼

Next Match

▲

Previous Match

| Score         | Expect                                                            | Method                       | Identities   | Positives    | Gaps      |
|---------------|-------------------------------------------------------------------|------------------------------|--------------|--------------|-----------|
| 221 bits(564) | 1e-72                                                             | Compositional matrix adjust. | 108/110(98%) | 109/110(99%) | 0/110(0%) |
| Query 1       | MALWMRLRLPL LALLALWGPDPAAAFVNQHL CGSHLVEALYLVCGERGFFYTPKTRREAE 60 |                              |              |              |           |
| Sbjct 1       | MALWMRLRLPL LALLALWGPDPAAAFVNQHL CGSHLVEALYLVCGERGFFYTPKTRREAE 60 |                              |              |              |           |
| Query 61      | LQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTISCLYQLENYCN 110             |                              |              |              |           |
| Sbjct 61      | LQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTISCLYQLENYCN 110             |                              |              |              |           |

Download

GenPept

Graphics

insulin [Pan paniscus]

Sequence ID: XP\_034787832.1 Length: 204 Number of Matches: 1

Range 1: 95 to 204

GenPept

Graphics

▼

Next Match

▲

Previous Match

| Score         | Expect | Method                       | Identities   | Positives     | Gaps      |
|---------------|--------|------------------------------|--------------|---------------|-----------|
| 224 bits(571) | 2e-72  | Compositional matrix adjust. | 109/110(99%) | 110/110(100%) | 0/110(0%) |

Related Information

Gene - associated gene details

AlphaFold Structure - 3D structure displays

Genome Data Viewer - aligned genomic context

Feedback

